Nathan Scott Johnson *Senior Scientist | Agentic AI, Autonomous Laboratories, and Materials Science*

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Professional Summary

Materials scientist and autonomous-systems researcher developing agentic AI for scientific instrumentation and self-driving laboratories. At Carl Zeiss Research Microscopy Solutions, I lead the development of LLM-controlled microscopy systems integrating planning, tool use, structured function calling, code generation, image analysis, verification, telemetry, and physical instrument control. My broader background spans high-throughput synthesis, active-learning-guided materials discovery, custom synchrotron instrumentation, and domain-specific evaluation of autonomous scientific systems.

Technical Expertise

Agentic AI and autonomous systems: LLM agents, multi-agent systems, long-horizon workflows, sequential decision-making, planning and tool use, structured function calling, code generation, workflow orchestration, verification, benchmark development, evaluation harnesses, telemetry, and failure analysis.

Software and machine learning: Python, scikit-learn, OpenCV, LangGraph, LangChain, Azure OpenAI, MCP/FastMCP, Ollama, Git, Bash, MATLAB, Gaussian-process regression and classification, active learning, computer vision, deep-learning based image segmentation.

Scientific automation: Instrument APIs, closed-loop experimentation, automated image analysis, simulator-based testing, scientific workflow generation, high-throughput experimentation, and integration of AI systems with real laboratory equipment.

Materials science: High-throughput synthesis, combinatorial sputtering, physical vapor deposition, alloy discovery, additive manufacturing, X-ray diffraction, X-ray tomography, SEM, EBSD, spectroscopy, and *in situ* characterization.

Experience

Carl Zeiss Research Microscopy Solutions — Dublin, CA

Senior Scientist, Materials Research — *Apr 2026 – Present*

Applications Development Scientist — *Jul 2023 – Apr 2026*

- Lead the development of autonomous, self-driving X-ray and electron microscopy systems that use large language models to operate scientific instruments, analyze experimental data, and complete multi-step materials-characterization workflows.
- Architect agentic systems incorporating planning, tool use, structured function calling, code generation, working memory, verification, and multi-agent orchestration for long-horizon scientific tasks.
- Build typed Python instrument and image-analysis tools using MCP/FastMCP, enabling agents to translate natural-language scientific objectives into executable microscope operations and analysis workflows.
- Develop domain-specific benchmark suites, test harnesses, and telemetry to quantify task success, reliability, execution time, resource use, and failure modes across agent architectures and system configurations.
- Validate autonomous workflows using instrument simulators and physical microscopes, with an emphasis on reproducibility, safe execution, and the practical limitations of probabilistic AI systems connected to laboratory hardware.
- Develop novel *in situ* X-ray and electron microscopy methods for engineering materials and collaborate with academic and industrial partners through publications, demonstrations, and international conference presentations.

Stanford University / SLAC National Accelerator Laboratory — Menlo Park, CA

Postdoctoral Scholar — *Mar 2021 – Jul 2023*

- Managed the research direction and operation of a high-throughput synthesis laboratory producing thin-film and bulk compositionally complex alloys.
- Integrated active learning and Bayesian experimental design with combinatorial synthesis and characterization workflows for accelerated materials discovery.
- Collaborated on the development of custom synchrotron instrumentation integrating X-ray diffraction, energy-dispersive X-ray spectroscopy, automation, and data analysis.

- Conducted research spanning thermoelectric alloys, pharmaceutical crystals, autonomous characterization, glass formation, and high-throughput materials experimentation.

Los Alamos National Laboratory — Los Alamos, NM
Graduate Research Assistant and ADAPT Fellow — *May 2018 – Jan 2021*
- Designed and implemented custom *in situ* X-ray diffraction platforms for studying additive-manufacturing processes and the mechanical response of complex 3D-printed structures.
- Researched machine-learning approaches for printable-alloy design and materials-process optimization.
- Collaborated with multidisciplinary teams across the Department of Energy on scientific instrumentation, additive manufacturing, synchrotron characterization, and materials modeling.

Education

Colorado School of Mines — Golden, CO
Ph.D. in Materials Science & Engineering — *2016 – 2021*
Advisor: Dr. Aaron P. Stebner. Developed novel *in situ* characterization methods for additive-manufacturing technologies.

The College of Wooster — Wooster, OH
B.A. in Physics, Minor in Mathematics — *2012 – 2016*
College Scholar Award; thesis received Honors.

Selected Publications

1. N. S. Johnson, "On the Optimization of Architecture, System Hyperparameters, and Context Databases for Agentic Self-Driving Microscopy," manuscript in preparation, 2026.
2. N. S. Johnson, "Multi-Agent Systems for Autonomous Laboratory Instrument Operation," preprint, 2024. Available: ResearchGate.
3. N. S. Johnson, et al., "Active Learning for Rapid Targeted Synthesis of Compositionally Complex Alloys," *Materials*, vol. 17, no. 16, p. 4038, 2024. Available: MDPI.
4. N. S. Johnson, et al., "Accelerating *In Situ* X-ray Tomography Using Sparse Projections and Deep Learning," *Materials Characterization*, article 116524, 2026. Available: ScienceDirect.
5. S. Huang, N. S. Johnson, et al., "Glass Formation During Combinatorial Sputtering in Binary Alloys," *Acta Materialia*, vol. 296, 2025. Available: ScienceDirect.

A full publication list is available on my Google Scholar profile.

Selected AI-for-Science Presentations

1. N. S. Johnson and I. Abshire, "Agentic Controllers for Self-Driving Microscopes and Automated Laboratories," MRS Spring Meeting, Honolulu, HI, Apr 2026.
2. N. S. Johnson, "Autonomous Control of an X-ray Microscope: A Multi-Agent Framework for Closed-Loop Materials Characterization," TMS Annual Meeting, San Diego, CA, Mar 2026.
3. N. S. Johnson, "An Autonomous, Large Language Model-Driven X-ray Microscope," AI for Materials Science, NIST, Gaithersburg, MD, Jul 2025.
4. N. S. Johnson, O. Hoidn, R. Tang-Kong, and A. Mehta, "Autonomous Spatial Mapping and Phase Detection in Low-Fidelity Samples Using X-ray Diffraction," Combinatorial Science, Golden, CO, 2022.

Selected Awards and Fellowships

AIME Champion H. Mathewson Award — **2024**
Recognized a significant contribution to the engineering application of metallic materials.

Los Alamos ADAPT Advanced Manufacturing Fellowship — **2018–2021**
Three-year graduate research fellowship supporting advanced-manufacturing and *in situ* characterization research.

Spring Commencement Student Speaker, Colorado School of Mines — **2021**
Selected as the Ph.D. commencement speaker following conference awards for research presentations.